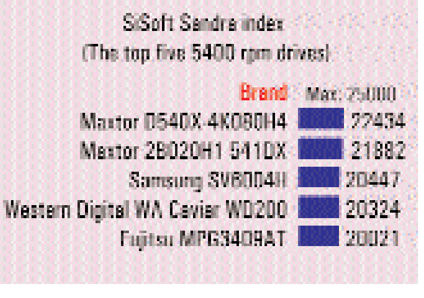
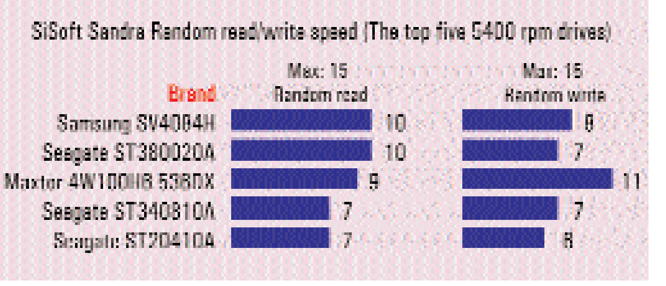
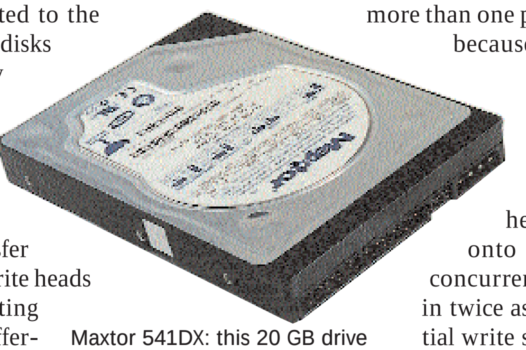
hard disk sub-system. The hard disk that did especially well in this test was the **Western Digital WA Caviar WD200**, clocking a very impressive score of 19900. This turned out to be higher than even two of the 7200-rpm drives. This implies that the hard disk, in spite of the 5400-rpm speed, is still capable of good performance in applications that require intensive access to the drive, such as multimedia authoring applications using audio and video. The Fujitsu MPG3409AT came a close second in this test, trailing slightly behind the Digital WA Caviar with a score of 18100. The drive that delivered a very low performance in this test was the 40 GB Seagate ST30810A, logging a score of just 7790. This implies that it isn't suited to situations where multiple applications would have to be executed at the same time. However, when it comes to storage space, this drive offers 40 GB and therefore imparts a lot of value in situations where you just need storage space, with little stress on performance.



In terms of the data transfer speeds, the drive that posted exemplary performance in the sequential read transfer rates was the **20 GB Maxtor 541DX**, which posted a score of 37 MBps. The drive that came in a close second here was its 80 GB sibling, the Maxtor D540X-4K080H4, clocking a respectable 33 MBps. The drive that trailed here was the 80 GB Seagate ST380020A, which clocked a very disappointing 14 MBps in the sequential read test. However, this drive posted a very respectable score of 10 MBps in the random read tests, along with the 40 GB Samsung SV4084H drive. This could be attributed to the fact that these hard disks also posted very good scores in the access time benchmarks. As access time has a direct bearing on the random data transfer rates, since the read-write heads are constantly actuating when accessing the different parts of the platters, a lower access time will help the heads access the data faster and subsequently allow for greater data transfer rates. These drives are ideally suited for applications where data will be primarily read in a random manner, for example, a storage drive where installations of applications are stored or databases are continually read from.



In the sequential writing tests, the drive that performed well was the **80 GB Maxtor D540X-4K080H4** and the 20 GB Western Digital Caviar WD200. In spite of them being 5400-rpm drives, they posted very respectable scores of 32 MBps and 29 MBps, respectively. An interesting trend noted here was that a majority of the drives that did well in the sequential write tests were those that featured more than one platter. This is because when there is more than one platter available, there are multiple heads writing onto the platters concurrently, resulting in twice as fast sequential write speeds. These drives would be ideally suited for applications such as amateur audio and video authoring, where you would be writing large quantities of data to the drive over sustained periods.



Maxtor 541DX: this 20 GB drive has a low CPU utilisation and has a fast sequential read rate

As seen from the graphs, the random write scores were expectedly logged as the lowest by these drives, since this test involved writing data on random sections of the hard disk.

The drive that did the best in this test was the mammoth **100 GB Maxtor 4W100H6 536DX**, which logged a good transfer rate of 11 MBps. Coming in second were the 40 GB drives, Fujitsu MPG3409AT and Samsung SV4084H.

How Much Space is Enough?

Application	Recommended capacity	Individual file size (Avg. range)	Explanation
Word processing, spreadsheets	At least 20 GB	20 KB to 200 KB	These files are very small and consume least amount of space on your drive. Go in for an entry-level 20 GB drive.
MP3 audio	20 GB and above, depending on the number of files	3 MB to 7 MB	The size of hard disk you settle for depends entirely on the total number of MP3 files you plan to store
Movies (DivX)	20 GB and above, depending on the number of movies	400 MB to 700 MB	Here again, the size of hard disk you settle for depends entirely on how many movies you need to store
Audio/video editing	Entry level: 20 GB Professional: 100 GB	Depending on usage	If you're a professional who works on audio/video files, you should opt for the largest and fastest drive you can afford.
Gaming	20 GB and above, depending on the number of games	Upwards of 20 MB	Today's games are huge and often take up to 1 GB on a full install. Depending upon whether you're a casual or serious gamer, you should go in for the largest drive you can afford